

Functional Specifications

Version	Date	Changes
1.0	2026/1/11	Initial specification

1 Description of Application

An online ridesharing system, where (1) riders and drivers can register the system, (2) drivers can advertise their planned routes (e.g., from PolyU to Central) in this system, (3) riders can view available routes, and (4) a rider can reserve a route for a shared ride.

1.1 Server Environment

Operating System: Windows, Linux or Mac OS

Web application infrastructure: Apache+PHP+MySQL, Node.js, Django, or equivalent framework

1.2 Coding Requirement

You are required to use Git/GitHub/Bitbucket or other code repository systems to host your codes to make it easy for version control and release. All codes should be properly refactored. That is, codes should be re-usable and readable at your best efforts. For example, any reusable codes should be encapsulated into functions and/or separate files for inclusion.

1.3 Client Environment

Client is a modern web browser, such as Chrome, Firefox or Internet Explorer.

2 List of Functions

Category	Deliverables
Rider/Driver Registration and Login	• Common information: login id, nick name, email, type (rider/driver) and profile image • Use id and password to login • Remember login status using cookie (login status remains even when the entire browser, not just the current tab, is closed) • User logout and delete cookie (login status is removed)
Driver Route Advertise	• A driver can advertise with his/her planned route, which includes date, time, starting location (text), destination (text), car model, available passenger capacity, and description
List of Available Routes	• A list page of all available routes in the system, each route has a “view” button
Route Detail	• A detailed page of a route, with a “reserve” button to reserve one seat in the route. If all capacity of this route is booked, the route becomes unavailable
List of My Booked Routes	• For rider: a list page of all reserved routes, each with an “cancel reserve” button to cancel the reservation. • For driver: a list page of all advertised routes, each showing a list of reserved passenger names.